QLAB · TJHSST

Recovering the Schwarzschild Metric

through Physics-Informed Neural Networks

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“Matter tells spacetime how to curve. Spacetime tells matter how to move.”

— John Archibald Wheeler

Einstein's general relativity describes gravity not as a force, but as the curvature of a 4-dimensional fabric called spacetime. Black holes are the most extreme example of that curvature.

Today's Roadmap

1. Black Holes & General Relativity
2. Tensors: The Language of GR
3. From Metric to Curvature
4. Deriving the Schwarzschild Metric
5. What is a PINN?
6. Our Network & Loss Function
7. Results & Future Work

PART 1

BLACK HOLES & GENERAL RELATIVITY

What we're actually trying to model

Spacetime, gravity, and the first exact solution Einstein's equations ever produced.

What Is a Black Hole?

A region where gravity is so strong

nothing escapes —

not even light.

Formed when a massive star collapses under its own gravity, crushing matter into a single point of (theoretically) infinite density — the singularity.

Singularity

Point of infinite density at the center

Event Horizon

Boundary of no return at r = 2M

Vacuum Outside

Empty spacetime, just curvature

Why Solve Einstein's Equations Numerically?

10

coupled nonlinear
partial differential equations

Einstein's full system

The problem

1 Exact solutions are rare

Only a handful of analytic solutions exist (Schwarzschild, Kerr, FLRW). Almost any realistic case is intractable by hand.

2 Numerical relativity is heavy

Existing codes are massive, specialized, and computationally expensive. Used for LIGO, but overkill for many problems.

3 Machine learning

A small neural network can learn metric components directly and are lightweight, mesh-free, easy to extend.

THE RESEARCH QUESTION

Can a physics-informed neural network recover the Schwarzschild metric directly from the vacuum Einstein equations — with no labeled solution data?

Why this is a strong test

Schwarzschild is simple enough to verify analytically, but rich enough to capture real nonlinear gravity. If a PINN can rediscover it, the method has potential for future extensions.

PART 2

TENSORS: THE LANGUAGE OF GENERAL RELATIVITY

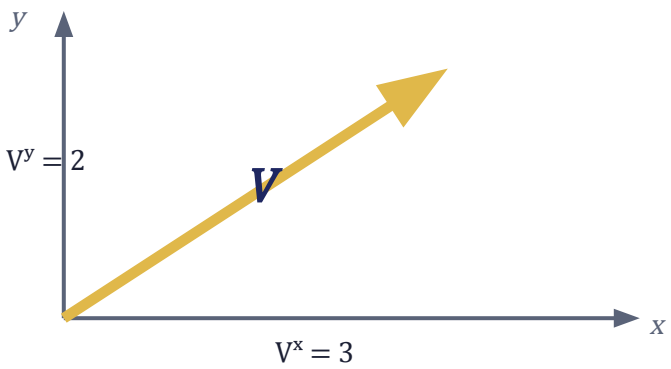
Vectors, components, and how geometry works

Everything in GR is built from tensors. We need to know what one is before we can do anything else.

Vectors

You are likely already familiar with these:

Arrow in space, with magnitude and direction:



In components

$$V = V^x e_x + V^y e_y$$

Three pieces:

components V^x, V^y — the numbers

basis vectors e_x, e_y — directions

vector V geometric object

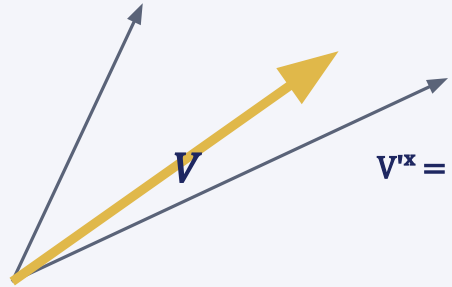
Components Change. The Vector Doesn't.

Frame A — standard axes



$V^x = 3,\ V^y = 2$

Frame B — rotated axes



$V'^x = 3.6,\ V'^y = 0.5$

Same arrow, different numbers. The components depend on your choice of frame; the vector itself does not.

How Components Transform: The Jacobian

$$V^{\alpha} = \Lambda_{\beta}^{\ \alpha}\ V^{\beta} \qquad \Lambda^{\alpha}\beta = \partial y^{\alpha} / \partial x^{\beta}$$

The new component

...is a weighted sum over all old components, with the Jacobian matrix Λ giving the weights.

The Jacobian

...is just partial derivatives of new coordinates with respect to old ones. A 4×4 matrix in spacetime.

THE RULE

An object whose components transform this way is called a

CONTRAVARIANT vector (upper index)

Examples: displacements dxμ, velocities, momenta

The Other Kind: Covariant Vectors (One-Forms)

CONTRAVARIANT

Vector Vμ

$$V^{\alpha} = \Lambda^{\alpha}{}_{\beta} V^{\beta}$$

Transforms with the Jacobian

Intuition: meters ⟶ cm,

a 2 m displacement becomes 200 cm.

Examples

velocity, momentum

COVARIANT (one-form)

One-form ωμ

$$\omega_{\alpha} = (\Lambda^{-1})^{\beta}_{\alpha} \omega_{\beta}$$

Transforms with the INVERSE Jacobian

Intuition: meters^{-1} ⟶ cm^{-1},

a gradient "per meter" becomes "per cm" 100× smaller.

Examples

gradients $\partial_{\mu} f$, the metric $g_{\mu\nu}$

The Metric Tensor: Measuring Spacetime

$$ds^2 = g_{\mu\nu}\, dx^\mu\, dx^\nu$$

The LINE ELEMENT — the squared distance between two nearby events.

FLAT (Minkowski)

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$$

$$g_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$$

Constant components. No gravity.

→ Just special relativity.

CURVED (general)

$$g_{\mu\nu} = g_{\mu\nu}(x)$$

Components depend on POSITION.

These components ARE the gravitational field.

→ What our network learns.

What Is a Tensor, Really?

A tensor is a machine with slots that eats vectors and one-forms, and produces a number.

Type	Eats	Indices	Example
Scalar (0,0)	nothing	$(none)$	Temperature T
Vector (1,0)	one one-form	V^{μ} — *one upper*	4-velocity u^{μ}
One-form (0,1)	one vector	ω_{μ} — *one lower*	Gradient $\partial_{\mu} f$
(0,2) tensor	two vectors	$T_{\mu\nu}$ — *two lower*	Metric $g_{\mu\nu}$, Ricci $R_{\mu\nu}$

Number of upper indices = how many one-forms it eats. Number of lower indices = how many vectors it eats.

Why Upper × Lower = Coordinate-Free

$$V^{\mu}\,\omega_{\mu} = V^{0}\omega_{0} + V^{1}\omega_{1} + V^{2}\omega_{2} + V^{3}\omega_{3} = a\ scalar$$

When an upper index meets a lower index of the same name,

we contract them — sum over all four values.

Why does this matter?

- V transforms with the Jacobian Λ.
- ω transforms with the INVERSE Jacobian Λ^{-1}.
- When multiplied together the two cancel exactly. The sum stays the same in every frame.
- **Physics is invariant. Our equations work in any coordinates.**

PART 3

FROM METRIC TO CURVATURE

Building the equations Einstein wrote down

Christoffel → Riemann → Ricci. The pipeline that turns the metric into a curvature law.

Specializing to a Black Hole: The Ansatz

$$ds^2 = -e^{2\alpha(r)}\,dt^2 + e^{2\beta(r)}\,dr^2 + r^2\,d\theta^2 + r^2\sin^2\theta\,d\varphi^2$$

1 Static

No time-dependence and no dt·dr cross-terms. Time-reversal symmetry kills them. So both unknowns depend only on r.

2 Spherically symmetric

The angular part must be the round 2-sphere: $r^2(d\theta^2 + \sin^2\theta\, d\varphi^2)$. The coefficient is absorbed into the definition of r.

3 Exponential form: why $e^{2\alpha}$, $e^{2\beta}$?

Guarantees positivity outside the horizon • simplifies derivatives • lets our network output any real α, β and stay well-defined.

The Curvature Pipeline



$$\text{Christoffel:}\quad \Gamma^{a}{}_{\nu\rho} = \tfrac{1}{2}\, g^{a\sigma}\left(\partial_{\nu} g_{\sigma\rho} + \partial_{\rho} g_{\sigma\nu} - \partial_{\sigma} g_{\nu\rho}\right)$$

$$\text{Riemann:}\quad R^{a}{}_{\sigma\mu\nu} = \partial_{\mu}\Gamma^{a}{}_{\nu\sigma} - \partial_{\nu}\Gamma^{a}{}_{\mu\sigma} + \Gamma^{a}{}_{\mu\lambda}\Gamma^{\lambda}{}_{\nu\sigma} - \Gamma^{a}{}_{\nu\lambda}\Gamma^{\lambda}{}_{\mu\sigma}$$

$$\text{Ricci:}\quad R_{\mu\nu} = R^{a}{}_{\mu a\nu} \ \text{(contraction of Riemann)}$$

Each step uses derivatives of the previous — that's why Einstein's equations are second-order and nonlinear.

Christoffel Symbols for Our Metric

Plugging the metric into the formula — the nonzero ones are:

$\Gamma^{t}_{tr} = \alpha'$	$\Gamma^{r}_{\theta\theta} = -\,r\,e^{-2I}$	$\Gamma^{\theta}_{\varphi\varphi} = -\sin\theta\cos\theta$
$\Gamma^{r}_{tt} = e^{2(a-I)}\,\alpha'$	$\Gamma^{r}_{\varphi\varphi} = -\,r\,e^{-2I}\sin^2\theta$	$\Gamma^{\varphi}_{r\varphi} = 1/r$
$\Gamma^{r}_{rr} = \beta'$	$\Gamma^{\theta}_{r\theta} = 1/r$	$\Gamma^{\varphi}_{\theta\varphi} = \cot\theta$

Worked example: deriving $\Gamma^{t}_{tr} = \alpha'$

$$\Gamma^{t}_{tr} = \tfrac{1}{2}\,g^{tt}\,\partial_r\,g_{tt} = \tfrac{1}{2}\,(-e^{-2a})(-2\alpha'\,e^{2a}) = \alpha'$$

(Static metric kills time derivatives; diagonal metric kills off-diagonal contributions.)

The Four Ricci Tensor Components

After contracting Riemann only 4 components are nonzero (and only 3 are independent):

$$R_{tt} = e^{2(\alpha-\beta)}\left[\alpha'' + (\alpha')^2 - \alpha'\beta' + 2\alpha'/r\right]$$

$$R_{rr} = -\alpha'' - (\alpha')^2 + \alpha'\beta' + 2\beta'/r$$

$$R_{\theta\theta} = 1 - e^{-2\beta}\left[1 + r(\alpha' - \beta')\right]$$

$$R_{\varphi\varphi} = \sin^2\theta \cdot R\theta\theta$$

Setting all four to zero (vacuum) gives us our ODE system.

Vacuum Einstein Equations

$$R_{\mu\nu} = 0$$

The vacuum Einstein equations, spacetime curvature where no matter is present.

1 Tensor equation

Holds in every coordinate system. Truly coordinate-independent physics.

2 Vacuum ≠ flat

Spacetime can still be curved, though not because of local matter. Schwarzschild proves this.

3 Underdetermined

Many solutions exist (Minkowski with M = 0, Schwarzschild with any M). We need a physical anchor to pick one.

Why $R_{\mu\nu} = 0$ in Vacuum

Step 1 — Full Einstein equation

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} = 8\pi T_{\mu\nu}$$

Step 2 — In vacuum, no matter: $T_{\mu\nu} = 0$

$$R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} = 0$$

Step 3 — Take the trace (multiply by $g^{\mu\nu}$)

$$R - \tfrac{1}{2} R\,(4) = -R = 0 \quad \Rightarrow \quad R = 0$$

Step 4 — Plug R = 0 back in :

$$R_{\mu\nu} = 0$$

PART 4

DERIVING THE SCHWARZSCHILD METRIC

Solving $R_{\mu\nu} = 0$ by hand

Three equations, three steps. We'll get the famous 1−2M/r factor that defines the black hole.

Step 1: Combine R_{tt} and R_{rr}

Multiply R_{tt} by $e^{-2(\alpha-\beta)}$ and add R_{rr} :

$$[\,\alpha'' + (\alpha')^2 - \alpha'\beta' + 2\alpha'/r\,] \;+\; [\,-\alpha'' - (\alpha')^2 + \alpha'\beta' + 2\beta'/r\,] \;=\; 0$$

$$\rightarrow \;\; 2\alpha'/r \;+\; 2\beta'/r \;=\; 0$$

$$\rightarrow \;\; \alpha' + \beta' \;=\; 0$$

$$\rightarrow \;\; \alpha(r) \;=\; -\,\beta(r) \;\; + \text{constant}$$

The constant can be absorbed by rescaling the time coordinate — a free choice. So we set α = −β.

Step 2: Solve $R_{\theta\theta} = 0$ with $\beta = -\alpha$

Substitute $\beta' = -\alpha'$ into $R_{\theta\theta} = 0$:

$$1 - e^{-2I}[\,1 + r(\alpha' - \beta')\,] = 0$$

$$1 - e^{-2I}[\,1 + 2r\alpha'\,] = 0 \quad (\text{since } \alpha' - \beta' = 2\alpha')$$

$$\text{Let } f(r) = e^{2a} = e^{-2I}. \text{ Then } f' = 2\alpha' \cdot f.$$

$$1 - f - r f' = 0$$

$$\rightarrow (r f)' = 1$$

$$\rightarrow r f = r + C \rightarrow f = 1 + C/r$$

Fix the integration constant by demanding Newtonian gravity at large r : $\varphi(r) \rightarrow -M/r$.

This forces $C = -2M$.

THE RESULT

The Schwarzschild Metric

$$e^{2a(r)} = e^{-2\lambda(r)} = 1 - 2M/r$$

$$ds^2 = -(1 - 2M/r)\,dt^2 + (1 - 2M/r)^{-1}\,dr^2 + r^2\,d\Omega^2$$

First exact solution to Einstein's field equations — Karl Schwarzschild, 1916.

Found within months of Einstein publishing GR, while Schwarzschild was serving on the Russian front in WWI.

Recap: What the Network Learns

In summary, the network needs to learn two radial functions $\alpha(r)$ and $\beta(r)$ such that:

Build metric	$g_{\mu\nu}$ has the spherical-ansatz form
Compute curvature	Auto-diff produces Γ, then Ricci $R_{\mu\nu}$
Force vacuum	R_{tt}, R_{rr}, $R_{\theta\theta}$ all evaluate to 0
Anchor mass	Recover M = 1, our chosen scale
Match analytic	$\alpha(r) = \frac{1}{2} \ln(1 - 2/r)$ with $M = 1$

PART 5

PHYSICS-INFORMED NEURAL NETWORKS

What's a PINN?

A neural network that learns to satisfy a differential equation with no training data at all.

Traditional Machine Learning vs. PINNs

TRADITIONAL ML

Learn from examples

Given thousands of input-output pairs, learn the mapping. Loss = mismatch with known answers.

Examples

Image classification · ChatGPT · spam filters

Problem for us

There's no training set of "Einstein equation solutions" to learn from.

PHYSICS-INFORMED NN

Learn from equations

Encode the PDE as the loss itself. Differentiate the network's output to evaluate residuals.

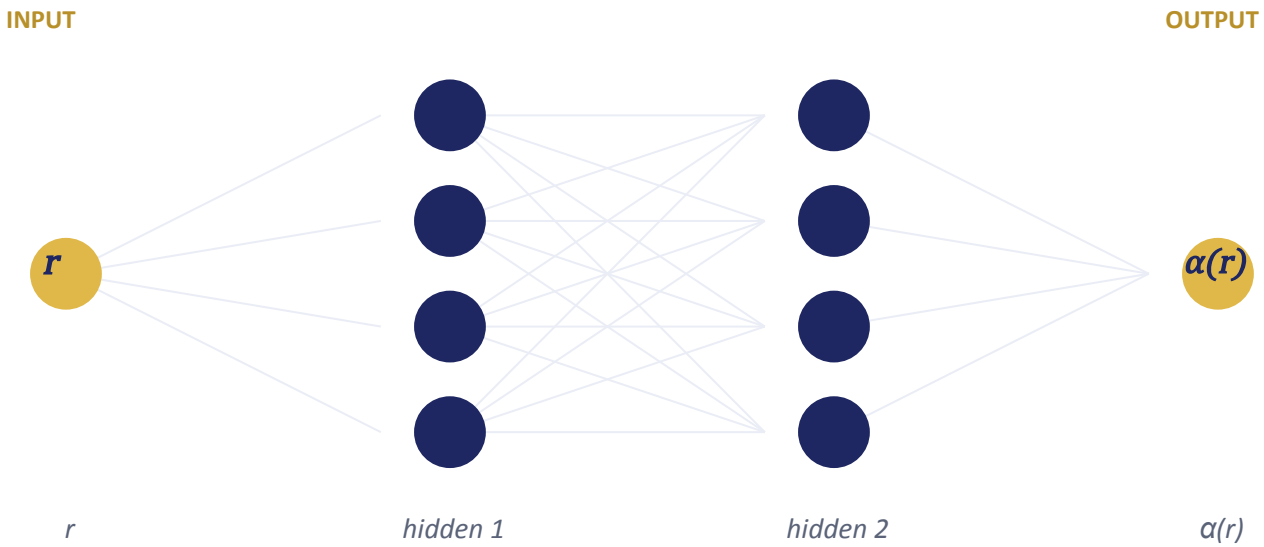
Examples

Fluid dynamics · quantum mechanics · general relativity

Why it fits

We HAVE the equations. We just need to satisfy them.

Quick Primer: What a Neural Network Is



A neural network is just a stack of weighted sums and nonlinear functions. Adjust the weights so the output approximates the function you want.

Automatic Differentiation

Modern ML frameworks compute exact derivatives through the network architecture.

If the network outputs α(r), we can compute:

$$\alpha'(r) = d\alpha/dr \quad \text{and} \quad \alpha''(r) = d^2\alpha/dr^2$$

exactly, in one line of code (using PyTorch).

1 Exact

Not approximated by finite differences. Fully analytic via chain rule.

2 Fast

Same cost as one forward pass. Built into the deep-learning frameworks.

3 Enables the loss

Lets us evaluate Ricci residuals which need second derivatives at every collocation point.

PART 6

OUR ARCHITECTURE & LOSS FUNCTION

How we built the network

Architecture choices that make training fast with a loss that picks out the correct black hole.

Our Network Architecture



Key design choice

Hard-code β = −α directly into the architecture.

This is the analytic relation we derived in Step 1 of the Schwarzschild derivation. By building it into the network instead of training for it, we halve the unknowns and make convergence dramatically faster.

The Loss Function: Three Pieces

$$L = \lambda_1 L_{PDE} + \lambda_2 L_{BC} + \lambda_3 L_{mass}$$

L_{PDE}

PHYSICS RESIDUAL

Sum of squared R□□, R_{rr}, $R_{\theta\theta}$ at sampled radial points.

Drives vacuum Einstein equations to zero.

L_{BC}

BOUNDARY CONDITION

Force $\alpha(r) \to 0$ as $r \to \infty$.

Asymptotic flatness, far away spacetime is Minkowski.

L_{mass}

PHYSICAL ANCHOR

Misner–Sharp mass equals 1.

Breaks degeneracy, selects Schwarzschild over flat space.

How Training Actually Works

1

Sample

Pick collocation points in r outside the horizon (logarithmically spaced).

2

Forward

Run the network to get α(r) at each sampled point.

3

Autodiff

Compute α′ and α″ via automatic differentiation.

4

Loss

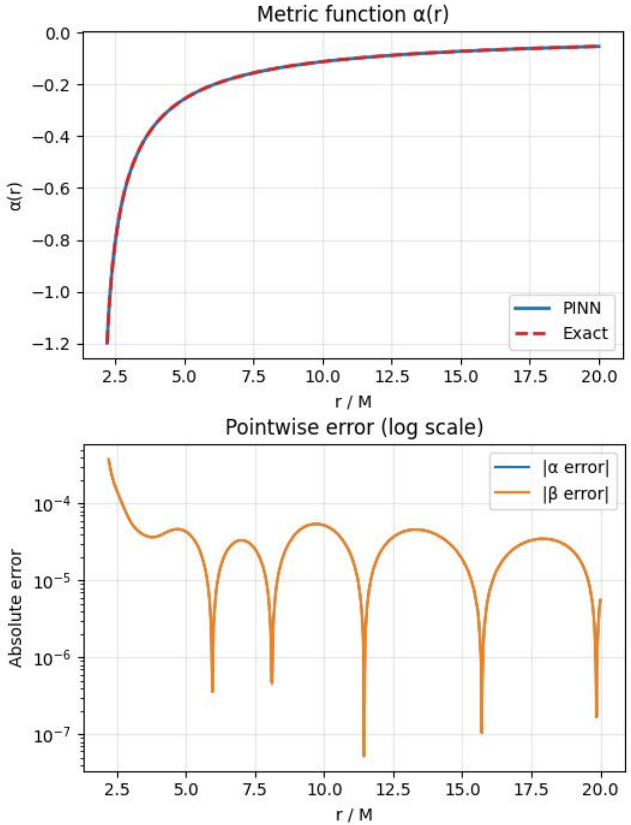
Build Ricci residuals + BC + mass loss and sum them.

5

Update

Backpropagate to nudge weights. Repeat until convergence to some minimum.

Results



$< 10^{-4}$

max relative error

vs. exact Schwarzschild

1

recovered mass M

Misner–Sharp constraint targeted M = 1

0

ground-truth points

trained only on equations + boundary

Results

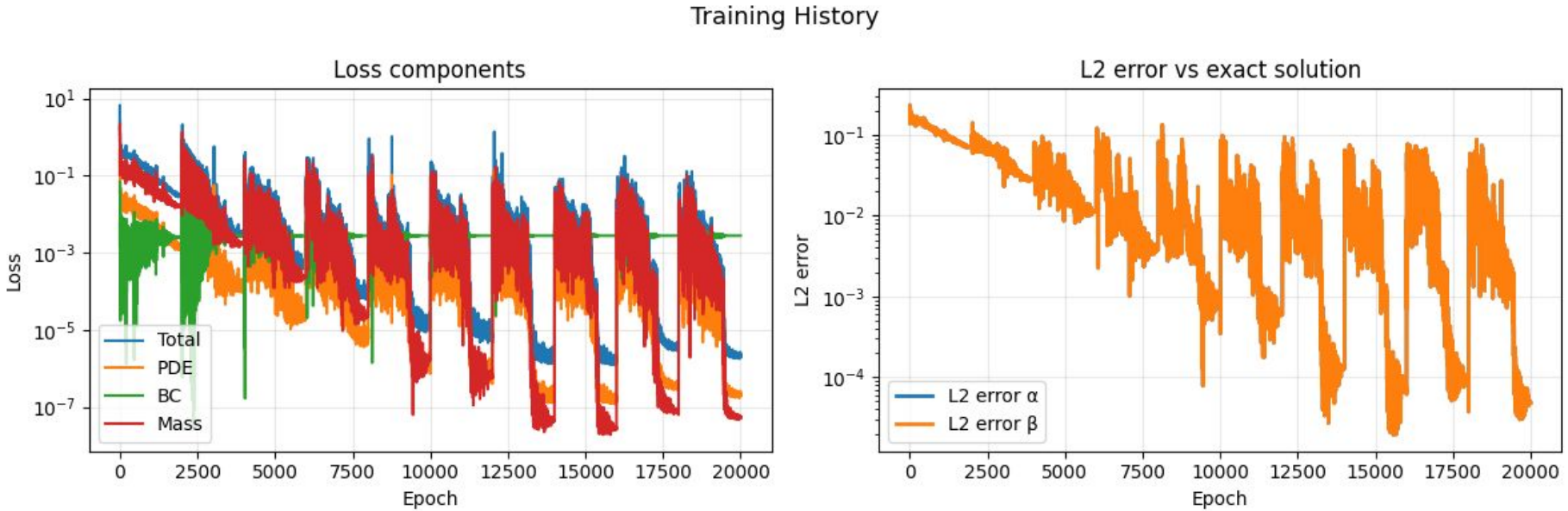
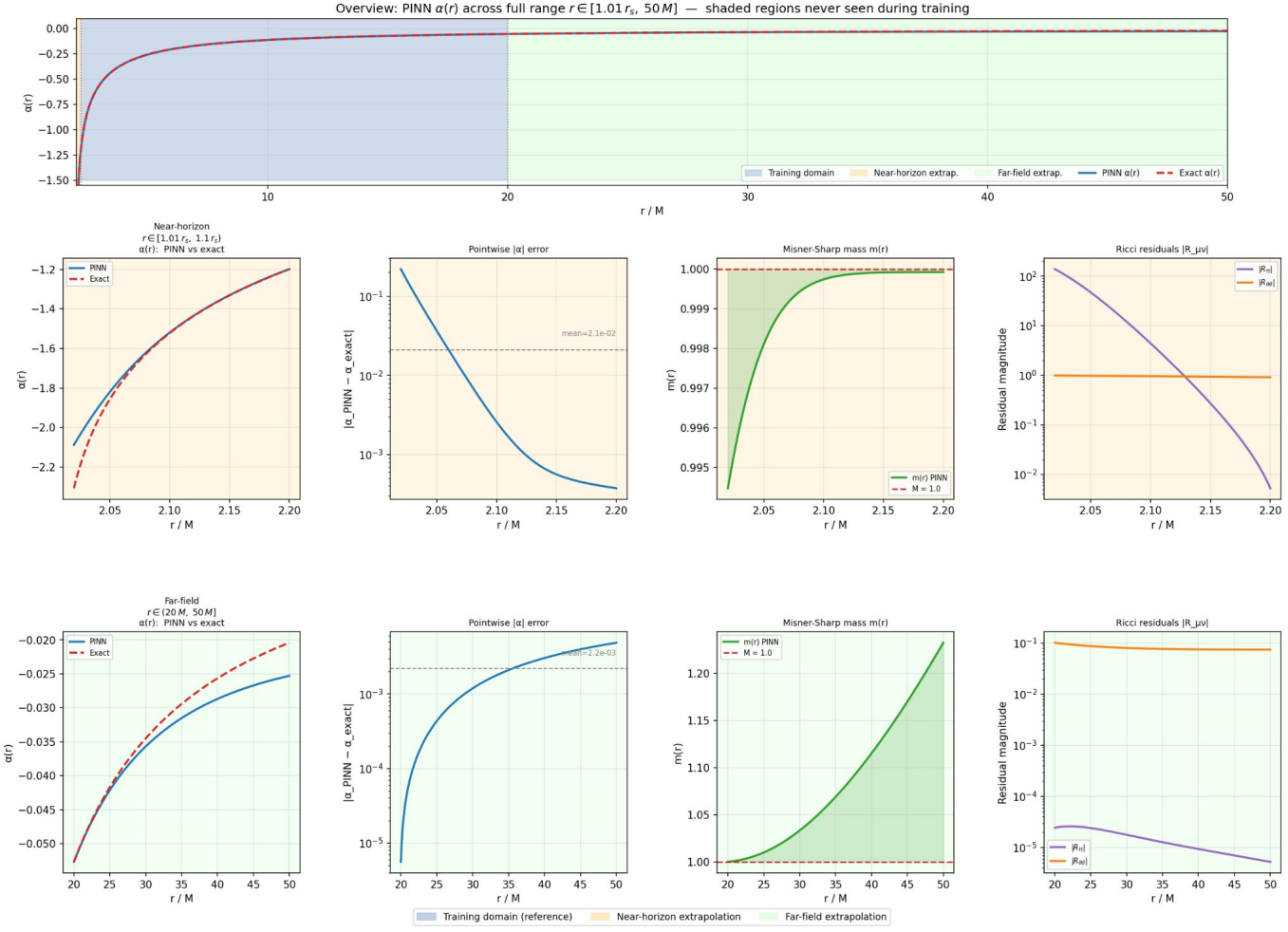

Training History
Loss components
Loss
Epoch
Total
PDE
BC
Mass
L2 error vs exact solution
L2 error
Epoch
L2 error α
L2 error β

Figure 3 — Extrapolation Diagnostic:
Physical Constraint Generalisation Beyond the Training Domain



Three Takeaways

01 PINNs can rediscover exact GR solutions

Neural networks can solve Einstein's equations from scratch with no labeled training data. The technique is generalizable.

02 Hard constraints beat soft penalties

Baking the analytic relation $\beta = -\alpha$ directly into the architecture was the single most effective design choice in reducing error. Whenever a physical relation is exact, encode it exactly.

03 Physical anchors are essential

Vacuum field equations are underdetermined. The Misner–Sharp mass constraint was what selected Schwarzschild over trivial Minkowski. Without it, the network wandered.

Where We Go Next

01

Kerr metric

Extend to 2D inputs (r, θ) for ROTATING black holes. Brings new cross-terms like $g_{t\varphi}$, a substantial architectural change.

02

Non-vacuum spacetimes

Introduce a stress-energy tensor $T_{\mu\nu} \neq 0$. Lets us model stellar interiors or black holes embedded in dark-matter halos.

03

Time-dependent backgrounds

Move past static metrics like ringdown after a merger, cosmological perturbations, gravitational radiation.

04

Mass loss as regularizer

Test whether keeping L_{mass} active throughout training (vs. early epochs only) acts as a permanent stabilizer.

THANK YOU

Questions?

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